



A molecular overview of the polymyxin-LPS interaction in the context of its mode of action and resistance development

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ABSTRACT

With the rising incidences of antimicrobial resistance (AMR) and the diminishing options of novel antimicrobial agents, it is paramount to decipher the molecular mechanisms of action and the emergence of resistance to the existing drugs. Polymyxin, a cationic antimicrobial lipopeptide, is used to treat infections by Gram-negative bacterial pathogens as a last option. Though polymyxins were identified almost seventy years back, their use has been restricted owing to toxicity issues in humans. However, their clinical use has been increasing in recent times resulting in the rise of polymyxin resistance. Moreover, the detection of “mobile colistin resistance (*mcr*)” genes in the environment and their spread across the globe have complicated the scenario. The mechanism of polymyxin action and the development of resistance is not thoroughly understood. Specifically, the polymyxin-bacterial lipopolysaccharide (LPS) interaction is a challenging area of investigation. The use of advanced biophysical techniques and improvement in molecular dynamics simulation approaches have furthered our understanding of this interaction, which will help develop polymyxin analogs with better bactericidal effects and lesser toxicity in the future. In this review, we have delved deeper into the mechanisms of polymyxin-LPS interactions, highlighting several models proposed, and the mechanisms of polymyxin resistance development in some of the most critical Gram-negative pathogens.

1. Introduction

The discovery and development of antibiotics were one of the most significant moments in the mid-twentieth century, often referred to as the “golden age of antibiotics” when several classes of antimicrobial substances were discovered and commercialized (Brown and Wright, 2016). However, the most serious consequence of widespread antibiotic use has been the emergence of drug-resistant bacterial strains which has motivated ongoing efforts to regulate the use of antibiotics. Bacteria have the remarkable ability to develop resistance to almost all antibiotics very shortly after their introduction, therefore investment in novel antibiotic discovery and development is a less rewarding prospect. The progressive decline in antibiotic discovery combined with the rapid emergence and spread of multi-drug resistant (MDR) bacteria presents an unprecedented challenge to the healthcare sectors worldwide (Davies and Davies, 2010).

Antimicrobial resistance (AMR) can be defined “as the ability of a microorganism to proliferate in the presence of an antimicrobial agent at a concentration that would ideally limit its growth” (Schrader et al.,

2020). AMR can arise due to several factors. Bacteria can be intrinsically resistant to several antibiotics, a feature of their genetic makeup, or can acquire resistance characteristics as well. Reduction in the outer membrane (OM) permeability and hyperactive efflux pumps are the attributes of intrinsic resistance mechanisms, whereas horizontal gene transfer (HGT) and acquisition of mobile genetic elements (MGE) are examples of acquired resistance mechanisms (Bhagirath et al., 2019). Categorically, Gram-negative bacteria owing to their typical cell membrane characteristics, exhibit more intrinsic resistance to several front-line drugs in comparison to their Gram-positive counterparts (Blair et al., 2015). For instance, the carbapenem-resistant *Enterobacteriaceae* (CRE), multi-drug resistant (MDR) *Pseudomonas aeruginosa*, and *Acinetobacter baumannii* are responsible for serious illnesses because of the limited therapeutic choices available to control them. Therefore, the World Health Organization (WHO) has included them in the priority pathogens list (PPL) against whom urgent therapeutic methods are needed (Tacconelli et al., 2018).

With the rising incidence of CRE, MDR *P. aeruginosa*, and MDR *A. baumannii*, and limited antibiotic choices against them, polymyxins

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remain the last line of drugs used (Vaara, 2019a). Polymyxins show rapid bactericidal activity against Gram-negatives, whereas, the Gram-positives, eukaryotic microorganisms, and human cells remain largely unaffected (HsuChen and Feingold, 1973). This selective toxicity against Gram-negative pathogens has made polymyxins a suitable class of antibiotics. Polymyxins primarily target the outer membrane (OM) that acts as a permeability barrier against several hazardous chemicals including antimicrobial substances. Polymyxin's interaction with the OM primarily occurs via electrostatic attraction between the cationic α , γ -di-aminobutyric acid (Dab) residues of polymyxin and anionic phosphate groups on the lipopolysaccharide (LPS) in the OM (Poirel et al., 2017b). Ca^{2+} and Mg^{2+} ions that bridge the neighboring LPS molecules to stabilize the structure are consequently dislodged, affecting the integrity of the membrane. This causes the OM to become more permeable, allowing intracellular materials to flow out causing cell death (Li et al., 2005; Aghapour et al., 2019). Polymyxins are primarily administered against the Gram-negative pathogens, still their increasing clinical use has resulted in the rise of polymyxin resistance.

Bacteria can develop mechanisms to circumvent stressful situations they encounter regularly in nature. Gram-negative pathogens can successfully adapt by changing the lipopolysaccharide (LPS) structure in the OM to protect themselves from cationic antimicrobial peptides (CAMPs), such as polymyxins. LPS modifications such as the attachment of substituents like 4-amino-4-deoxy-L-arabinose (L-Ara4N), phosphoethanolamine (pEtN), or galactosamine to the lipid A results in the overall decrease in the negative charges of phosphate residues. Consequently, the modified LPS shows diminished binding affinity to polymyxins on the membrane surface limiting their further entry into the cell (Olaitan et al., 2014; Sabnis and Edwards, 2023).

The intricacies of the LPS modification pathway are centered around the bacterial two-component signalling systems (TCS). To maintain homeostasis, TCSs, which are present in nearly all bacterial species, enable the bacteria to sense their surroundings and respond appropriately (Tierney and Rather, 2019). The canonical bacterial TCS comprises two major elements; a sensor protein that also is an autophosphorylating histidine kinase, and a response regulator that doubles as a transcription factor (Capra and Laub, 2012; Zschiedrich et al., 2016). Detecting an extracellular stimulus, the cell membrane located sensor kinase gets auto-phosphorylated at a conserved histidine residue and subsequently transfers the phosphate to a conserved aspartic acid residue on the cognate response regulator protein in the cytoplasm. The activated response regulator then modulates the expression of the target genes. Many of the genes having a role in LPS modification are controlled by TCSs. Therefore, mutations in the TCS-encoding genes may lead to the constitutive overexpression of LPS-modifying genes resulting in polymyxin resistance (Huang et al., 2020). Apart from the adaptive mechanisms responsible for the rise of polymyxin resistance, the discovery of the mobile colistin resistance genes (*mcr*) carried on plasmids has been detected across the globe (Liu et al., 2016; Wang et al., 2018).

Polymyxins though were discovered in the 1950 s, their use in the clinic was restricted due to toxicity issues. However, with the availability of comprehensive pharmacokinetics data, and our increasing understanding of their mechanism of action, there is an upward trend in their use in the healthcare sector as a last option antibiotic. Unfortunately, this has resulted in increasing reports of polymyxin resistance in healthcare settings. In this context, a molecular-level understanding of the polymyxin's mode of action at the bacterial OM for developing novel versions of the drugs as well as designing better delivery methods into the sites of infection is paramount. In this article, we have attempted to present the recent developments in the area to understand polymyxin's mechanism of action and pathways for resistance development concerning a few important nosocomial pathogens. Moreover, the article highlights some of the recent molecular interaction studies involving bacterial membranes and polymyxins that have helped in our understanding of the subject in molecular detail.

2. Polymyxins- a structural overview

Polymyxins belong to a group of antimicrobial substances that are synthesized as non-ribosomal cyclic peptides in the Gram-positive soil bacterium of the genus *Paenibacillus*. Benedict and Langlykke (1947) discovered polymyxin from *Paenibacillus polymyxa* previously known as *Bacillus polymyxa* var. *colistinus*. Based on their structural variations fifteen types of polymyxins are known of which polymyxin B and colistin (polymyxin E) are the most significant ones. Polymyxins are cationic lipopeptides of about 1.2 kDa in mass and consist of a heptapeptide ring structure with a linear tripeptide fragment acylated by fatty acid residues at the N-terminus of the tripeptide chain (Fig. 1A). Polymyxins have a common structural feature that includes L-di-aminobutyric acid (Dab), threonine, and a branched fatty acid (Trimble et al., 2016).

The amphoteric nature of the polymyxins can be attributed partially to the hydrophilic peptide ring, which is cationic, and the hydrophobic fatty acyl chain. This structural feature is necessary for their antibacterial action. Structurally, the primary amino acid sequences of both polymyxin B and colistin are similar, with a few notable exceptions, where D-phenylalanine in polymyxin B is substituted by D-leucine in colistin (Fig. 1A) (Velkov et al., 2013). These structural features are also the reasons for polymyxins showing nephrotoxicity and neurotoxicity toward human cells (Poirel et al., 2017b).

3. Gram-negative cell envelope- the primary target of polymyxins

Gram-negative bacteria are referred to as “di-derms”, because of the presence of a typical dual membrane system consisting of a cytoplasmic inner membrane (IM) and an OM. Like most biological membranes, the IM is a typical phospholipid bilayer, whereas the OM is asymmetric, comprising a phospholipid inner leaflet and a lipopolysaccharide (LPS) outer leaflet (Sperandeo et al., 2019). The OM is the initial point of contact with the surroundings, so its maintenance is essential for bacterial survival. Alterations in this structure impact the basic bacterial activities including motility, adhesion, and pathogenicity. The hydrophobic nature of the lipid bilayer of OM hinders the passage of polar molecules by electrostatic repulsion (Rosas and Lithgow, 2022). The structural characteristics of the LPS molecule help in protecting Gram-negative bacteria from severe environmental perturbations as well as CAMPs. LPS, particularly its lipid A moiety, is involved in pathogenesis by influencing host-pathogen interactions (Steimle et al., 2016). In the following sections, the structural overview of the LPS and its synthesis is presented.

3.1. Structure of lipopolysaccharides (LPS)

The LPS molecule, also known as endotoxin, is constituted of lipid A, the core polysaccharide, and the O- polysaccharide also known as the O-side chain or O-antigen (Samantha and Vrieling, 2020). Lipid A contains two glucosamine (D-GlcN) sugar derivatives, each with fatty acids and phosphates attached. The core polysaccharide contains various hydrophilic anionic groups, 3-deoxy-d-manno-2-octulosonic acid (Kdo) in the core region along with L-glycero-D-manno-heptose (L,d-Hep) and several hexose sugar moieties attached to the lipid A covalently. Among the Gram-negative organisms, the core area is connected to the O-antigens, which are repeating units of sugars (Sperandeo et al., 2019; Sabnis and Edwards, 2023) (Fig. 1B).

3.2. LPS biosynthesis pathway

Production of full-length LPS molecules is dependent on the coordination of several independent pathways that take place in various cellular niches such as the cytoplasm, cytoplasmic membrane, and periplasm. This mechanism is well conserved and has been studied in detail in *E. coli* and *Salmonella*. The initial step of the LPS biosynthetic

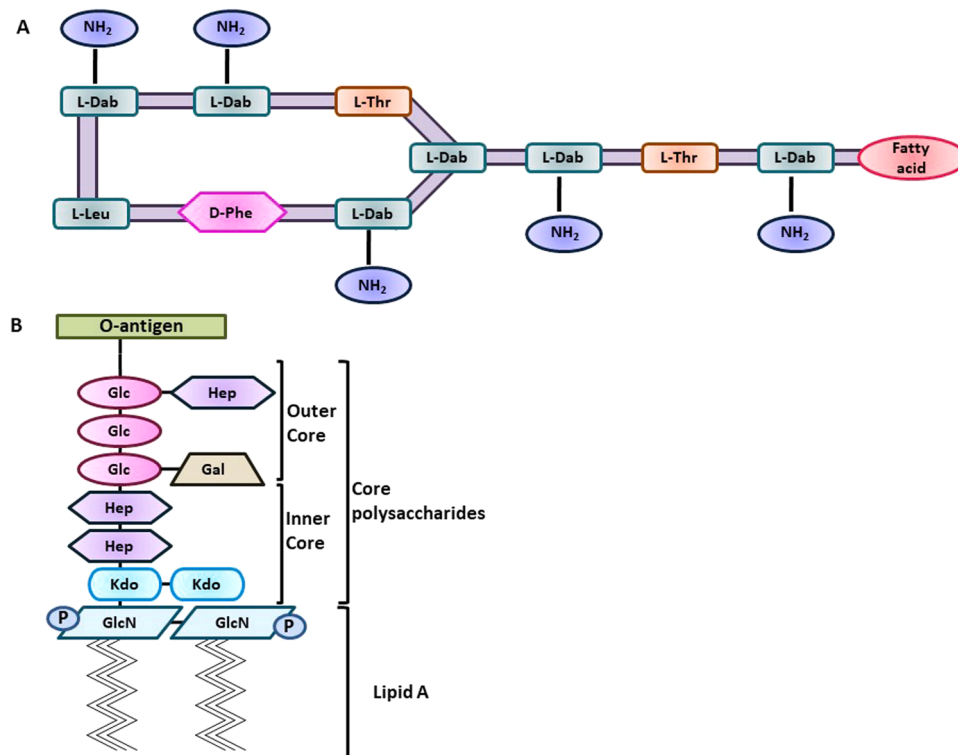


Fig. 1. (A) General structure of polymyxin. Polymyxins consist of L-di-aminobutyric acid (L-Dab), the amino acid threonine (L-thr), and a branched-chain fatty acid. The amino acid D-phenylalanine is present in polymyxin B but in the case of colistin, it is substituted by D-leucine. (B) The structural components of lipopolysaccharide (LPS). LPS is composed of three parts: Lipid A, Core polysaccharide, and O-antigen. Two glucosamine (D-GlcN) sugar derivatives are present in lipid A linked with phosphates and fatty acids. Lipid A is attached to an inner core, which contains sugars such as 3-deoxy-d-manno-2-octulosonic acid (Kdo) and L-glycero-D-manno-heptose (Hep). The outer core contains common hexose sugars such as glucose (Glc) and galactose (Gal). The O-antigen region of the LPS molecule is made up of repeated units of common sugars. (See the text for details).

pathway known as the “Raetz Pathway”, involves the synthesis of the Kdo2-lipid A structure, the details of which were deciphered by Christian Raetz and colleagues (Bertani and Ruiz, 2018) and presented in Fig. 2. The Raetz Pathway consists of nine enzymatic steps that take place in the cytoplasm and the inner leaflet of the IM (Raetz and Whitfield, 2002; Raetz et al., 2007, 2009). Interestingly, this pathway is absent in some specialized endosymbionts such as *Wolbachia* (Foster et al., 2005) and *Borrelia burgdorferi* that causes Lyme disease (Takayama et al., 1987) due to the lack of lipid A biosynthetic (*lpx*) genes, and organisms like *Sphingomonas* species possessing glycosphingolipids instead of lipid A (Kawahara et al., 2000). This is useful for distinguishing different bacterial species and determining whether bacteria are rough (R) or smooth (S) types (Mazgaen and Gurung, 2020). LPS having all three components is called the S form, whereas bacteria like *Neisseria meningitidis* and *Bordetella pertussis*, lacking the O-antigen are referred to as the R form or simply lipooligosaccharide (LOS) (Putker et al., 2015).

The biosynthesis of Kdo2-lipid A in *E. coli* begins with the acylation of the molecule, UDP-N-acetylglucosamine (UDP-GlcNAc) by the acyltransferase enzyme LpxA, producing UDP-3-O-(acyl)-GlcNAc, which is further deacetylated to UDP-3-O-(acyl)-GlcN, a reaction catalyzed by the Zn²⁺-dependent deacetylase enzyme LpxC (Jackman et al., 1999; Bertani and Ruiz, 2018). The first step, involving the LpxA enzyme, is unfavorable, hence the following deacetylation step catalyzed by LpxC is the “first committed step” of Kdo2-lipid A synthesis. After deacetylation, the acyltransferase enzyme LpxD produces UDP-2,3-diacylglucosamine which is a non-reducing sugar precursor of the lipid A backbone (Fig. 2). The pyrophosphate link is cleaved, resulting in the production of 2,3-diacylglucosamine-1-phosphate (also known as lipid X) in presence of a pyrophosphatase enzyme LpxH. This step is catalyzed by various non-homologous enzymes. For example, most *gamma*- and *beta*-

proteobacteria use the LpxH enzyme, whereas LpxI catalyzes the reaction in *alphaproteobacteria* and LpxG, having homology to both LpxH and LpxI, participates in the *Chlamydiae* species. In the next step, lipid X is linked with UDP-2,3-diacylglucosamine by a β -1'-6 glycosidic bond to produce lipid A disaccharide, catalyzed by LpxB releasing a molecule of UDP. The lipid A disaccharide is phosphorylated by the kinase LpxK to produce lipid IV_A. Following that, WaaA (formerly KdtA), a bifunctional enzyme, adds Kdo residues to lipid IV_A to yield the Kdo₂-lipid IVA molecule (Fig. 2). The LPS biosynthesis concludes with the addition of the secondary lauryl and myristoyl units to Kdo₂-lipid IV_A, resulting in a final Kdo₂-lipid A product, a reaction catalyzed by two acyltransferase enzymes LpxL and LpxM. The lipid A-oligosaccharide core (LOS) once assembled at the cytoplasmic face of the IM, MsbA, an ABC transporter, flips this molecule onto the periplasmic face of the IM (Simpson and Trent, 2019; Sabnis and Edwards, 2023).

In O-antigen biosynthesis, the repeated units are generated in the cytoplasm and attached to the undecaprenyl pyrophosphate, a lipid carrier in the presence of a Wzy-dependent O-antigen pathway. Then this complex structure is flipped to the periplasmic face of the IM via the O-antigen flippase transporter. The lipid A-core (LOS) and O-antigen biosynthesis routes converge on the IM at the periplasmic face, and both structures are ligated by the enzyme WaaL to produce a mature LPS molecule (Han et al., 2012; Ruan et al., 2012; Bertani and Ruiz, 2018). Subsequently, the LPS or LOS is carried to the OM surface by the Lpt (LPS transport) system. The Lpt pathway consists of seven proteins that include an IM complex i.e., LptB₂FG that extracts LPS from the IM, and this complex interacts with LptC, an LPS-binding protein, to transfer LPS from the IM's periplasmic leaflet to LptA that bridges the IM and OM and allows the LPS molecule to cross the periplasm, and an OM complex i.e., LptDE that allows the large LPS molecule to cross the OM through the outer leaflet for its final destination (Zhang et al., 2013; Bertani and

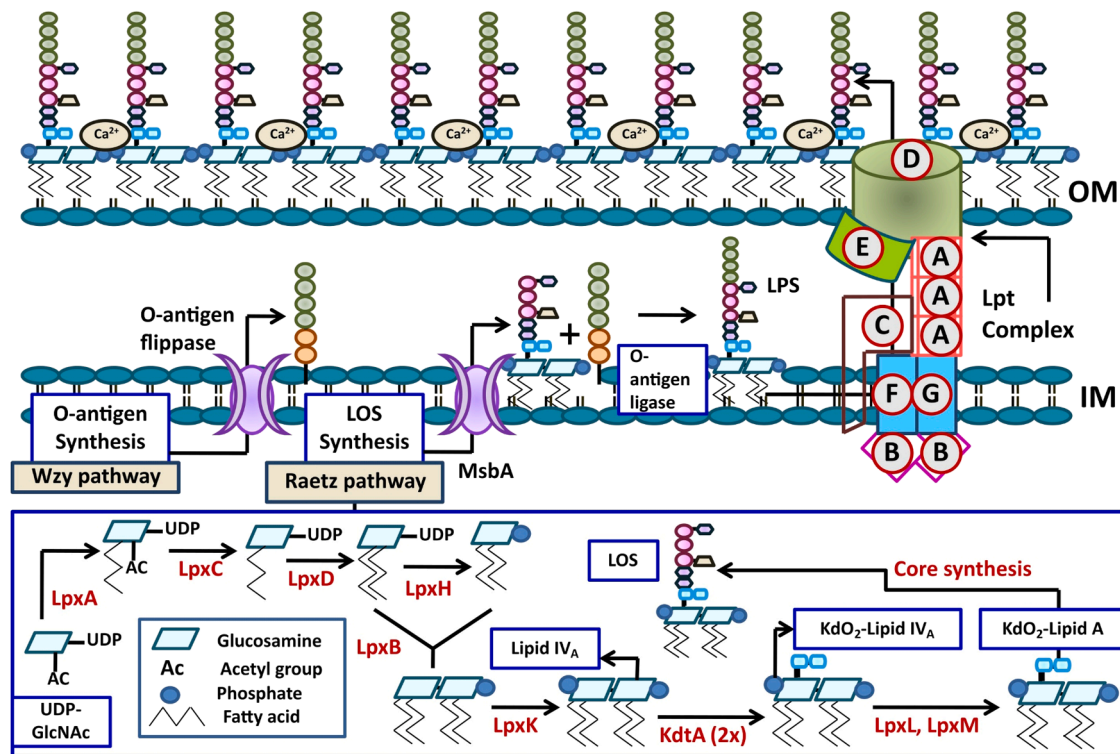


Fig. 2. Biosynthesis of lipopolysaccharides (LPS). The nine-step Raetz pathway synthesizes the lipid A and core domains of LPS in the cytoplasm, whereas the Wzy pathway assembles O-antigen independently in the cytoplasm. Lipooligosaccharide (LOS) is flipped across the inner membrane by MsbA, an ABC transporter, whereas O-antigen flippases flip O-antigen molecules. O-antigen is connected to LOS on the periplasmic side of the inner membrane. LPS is subsequently transported from the inner membrane to the outer membrane's surface via the Lpt complex. (See the text for details).

Ruiz, 2018; Sabnis and Edwards, 2023) (Fig. 2).

4. Mechanisms of polymyxin bactericidal activity

Polymyxins being amphipathic have a unique mode of action responsible for their very efficient bactericidal activity. Polymyxins generate a disruptive effect on bacterial membranes leading to permeability changes causing cell death. Though recent studies have revealed quite a few details of polymyxin's mode of action, specifically its molecular underpinnings, a thorough understanding is necessary for designing various polymyxin derivatives and developing novel treatment regimens using them. The different models proposed for the polymyxin antibacterial activity have been reviewed previously (Poirel et al., 2017b; Mohapatra et al., 2021) and briefly summarized in the following sections (Fig. 3).

4.1. Membrane disruption model

Polymyxins primarily target the membranes of Gram-negative bacteria, and the mechanism is explained by a long-accepted model known as the "self-promoted uptake" pathway. This model takes into consideration that the amphipathicity of polymyxin is essential for its absorption through the OM barrier (Kaye et al., 2016), where the cationic di-aminobutyric acid (Dab) residues of polymyxin interact electrostatically with anionic phosphate groups on lipid A (Fig. 3). This interaction destabilizes the LPS structures that are held together by the cations Mg^{+2} and Ca^{+2} . The D-Phe⁶-L-Leu⁷ of polymyxin B (D-Leu⁶-L-Leu⁷ in colistin) along with the hydrophobic N-terminal fatty acyl chain regulate their entry into the hydrophobic region of the OM. This interaction creates destabilized areas where polymyxin can flow through OM by weakening the packing of adjacent lipid A fatty acyl chains and damaging the IM phospholipid bilayer. This leads to increased membrane permeability, followed by the leaking of cellular contents, causing cell lysis (Li et al.,

2005; Nang et al., 2021).

Though the primary target of polymyxin remains bacterial OM, however, it is not enough to cause sufficient cell damage. Recently, Sabnis et al. (2021) have convincingly demonstrated that colistin targets the nascent LPS too in *E. coli* IM disrupting its structure. Furthermore, the study also reported that artificially increasing the concentrations of LPS in the IM (by adding the antibiotic murepavadin) makes the cell more susceptible to colistin which was further corroborated in a mice model infected with *P. aeruginosa* (Sabnis et al., 2021), and a subsequent study too highlighted the hitherto underappreciated role of IM in the polymyxin resistance (Humphrey et al., 2021). Another study using sophisticated atomic force microscopy (AFM) has explored the polymyxin-LPS interaction using the outer membrane vesicles (OMV) (Manioglu et al., 2022). The interaction of polymyxin when used at a concentration corresponding to the MIC with the OMVs showed up typical crystalline structures with regular hexagonal lattices in the membrane. However, such structures were not found in a strain carrying *mcr-1* plasmid that confers polymyxin resistance. This indicated that the crystalline structures in the membrane when treated with polymyxin may be a mechanism contributing to polymyxin's mode of action. Specifically, it is suggested that the formation of the crystalline structure leads to an increase in the membrane stiffness that can be correlated with the appearance of membrane bulges on bacterial cell surface when treated with polymyxins (Manioglu et al., 2022).

Recently, many analytical methods have been used to investigate the interaction of LPS with polymyxin which is extensively covered in Section 6.

4.2. Vesicle-vesicle contact pathway

The vesicle-vesicle contact is another bactericidal mechanism, in which polymyxin binds to the phospholipids in both the IM and OM causing the molecules to exchange between the membranes (Fig. 3). The

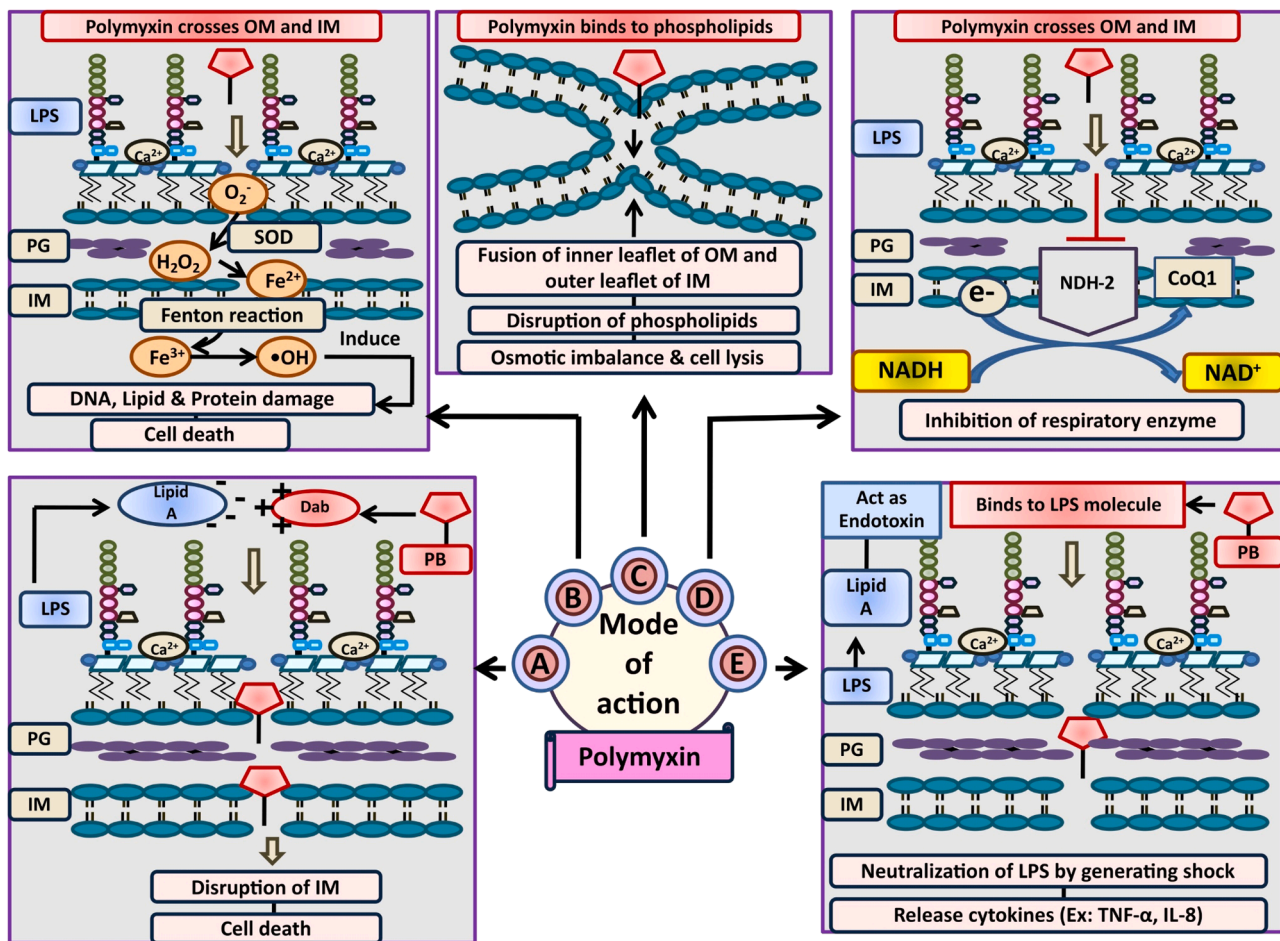


Fig. 3. Different modes of action of polymyxin. (A) Membrane disruption model; (B) Hydroxyl radical death pathway; (C) Vesicle-vesicle contact pathway; (D) Inhibition of respiratory enzymes; and (E) Anti-endotoxin activity. (See the text for details).

phospholipid composition is disrupted because of this process, causing osmotic imbalance and cell lysis (Cajal et al., 1995, 1996; Clausell et al., 2007; Yu et al., 2015).

4.3. Hydroxyl radical death pathway

Reactive oxygen species (ROS), such as hydroxyl radicals ($\bullet\text{OH}$), superoxide (O_2^-), and hydrogen peroxide (H_2O_2) produced as a result of the oxidative stress, also contribute to the polymyxin antibacterial activity and the pathway known as the “hydroxyl radical death pathway”. When polymyxin enters the bacterial cell envelope, O_2^- is produced, which is further converted to H_2O_2 , catalyzed by superoxide dismutase (SOD). H_2O_2 not only oxidizes Fe^{2+} to Fe^{3+} but also helps in producing $\bullet\text{OH}$ and the reaction is known as the “Fenton reaction” (Fig. 3). With the $\bullet\text{OH}$ concentration reaching a critical level, it causes oxidative damage to DNA, lipids, and proteins, resulting in cell death (Kohanski et al., 2007; Imlay, 2015; Yu et al., 2015; Rhouma et al., 2016).

4.4. Respiratory enzyme inhibition

Polymyxin’s secondary mechanism of action involves the interference in the functioning of type II NADH-quinone oxidoreductases (NDH-2) involved in respiration (Fig. 3). Polymyxin blocks NADH oxidase enzymes in Gram-positive *Bacillus* spp., as well as an alternate NADH-dehydrogenase and malate: quinone oxidoreductase in *Mycobacterium smegmatis* (Tochikubo et al., 1986; Mogi et al., 2009; Yu et al., 2019). This inhibitory mechanism seems to be widespread in the bacterial world as *E. coli*, *K. pneumoniae*, and *A. baumannii* also possess it (Deris

et al., 2014a).

4.5. Anti-endotoxin activity

The lipid A can impart a strong immune response and therefore it is referred to as an endotoxin. Hence with binding to lipid A, the polymyxins show strong anti-endotoxin activity. This binding inhibits LPS endotoxin action to generate shock by releasing cytokines like tumor necrosis factor- α (TNF- α) and interleukin 8 (IL-8) into the body (Fig. 3). This phenomenon is not completely understood as of now considering that various LPS-binding proteins in the plasma can bind with the endotoxin to neutralize its effect (Gough et al., 1996; Falagas and Kasiakou, 2005).

5. Polymyxin resistance mechanisms

Polymyxins are drugs of last choice against MDR pathogens; however, the increasing reports of polymyxin-resistant strains across the globe have seriously limited their therapeutic value, warranting understanding the pathways contributing to the rise of polymyxin resistance. Interestingly, the Gram-negative bacterial species such as *Proteus mirabilis*, *Moraxella catarrhalis*, *Helicobacter pylori*, *Serratia marcescens*, *Providencia* spp., *Burkholderia cepacia*, *Edwardsiella tarda*, etc. are intrinsically resistant to polymyxins. These species have their LPS modified by the addition of cationic groups such as phosphoethanolamine (pEtN) and amino arabinose (L-Ara4N) resulting in the repulsion of cationic polymyxins (Raetz et al., 2007; Olaitan et al., 2014; Simpson and Trent, 2019). However, polymyxin resistance can develop or be

acquired by the sensitive bacterial species by distinct modes such as - (i) chromosome-encoded resistance (via spontaneous mutagenesis) and (ii) transposon/plasmid-mediated transferable resistance (horizontally acquired resistance) (Poirel et al., 2017b; Mohapatra et al., 2021) which are discussed in the following sections. The basic resistance mechanisms are essentially similar among various Gram-negative species, albeit with subtle differences among them. Our current understanding of the molecular mechanisms involved in polymyxin resistance is presented in the following sections as well as illustrated in Fig. 4.

5.1. Chromosome-encoded resistance to polymyxins

Almost all bacterial species have evolved special strategies to survive in stressful situations. Among the bacterial species showing, intrinsic or acquired, resistance to polymyxins the following mechanisms have been described; (i) LPS modification via the addition of cationic groups, (ii) complete loss of the LPS, (iii) upregulation of efflux pump systems, (iv) capsule formation, and (v) enzymatic degradation of colistin, etc. (Poirel et al., 2017b; Yin et al., 2019; Lee et al., 2022). In this review, we have aimed to provide an overview of the mechanisms underlying the development of polymyxin resistance, with a focus on the alteration/modification of the LPS molecule. Most of the polymyxin resistance strategies involve modification of the LPS structure leading to a decreasing interaction with the cationic polymyxin molecule.

The addition of cationic moieties such as phosphoethanolamine (pEtN), 4-amino-L-arabinose (L-Ara4N), and galactosamine to lipid A is the most common LPS modification observed among bacterial pathogens (Baron et al., 2016). As a result, the net negative charges on the LPS are decreased influencing the electrostatic contact with polymyxin (Jiang et al., 2020).

The biosynthesis and addition of L-Ara4N depend on the coordination of several enzymes such as PmrE, PmrH, PmrF, PmrI, PmrJ, PmrK, PmrL, and PmrM (also known as Ugd, ArnB, ArnC, ArnA, ArnD, ArnT, ArnE, and ArnF, respectively) (Gunn et al., 1998; Yan et al., 2007), encoded by the *arnBCADTEF* operon (also called the *pmrHFIJKLM* operon) and are regulated by various two-component signalling (TCS) systems (Poirel et al., 2017b; Mohapatra et al., 2021). The L-Ara4N once produced in the cytoplasm moves to the IM, specifically to its outer leaflet, followed by getting transported to the lipid A (Trent et al., 2001; Yan et al., 2007) (Fig. 4). The L-Ara4N modification is more effective than the phosphoethanolamine (pEtN) modification because the former reduces the net charge of lipid A to zero, whereas the latter decreases it from -1.5 to -1 . The binding capacity of polymyxins with the modified LPS molecule is reduced because of the decreased overall negative charge (Nikaïdo, 2003; Olaitan et al., 2014).

The genes involved in the LPS alteration pathways are directly regulated by various TCSs. Therefore, mutations in the TCS encoding genes, often arising spontaneously, influence their expression and

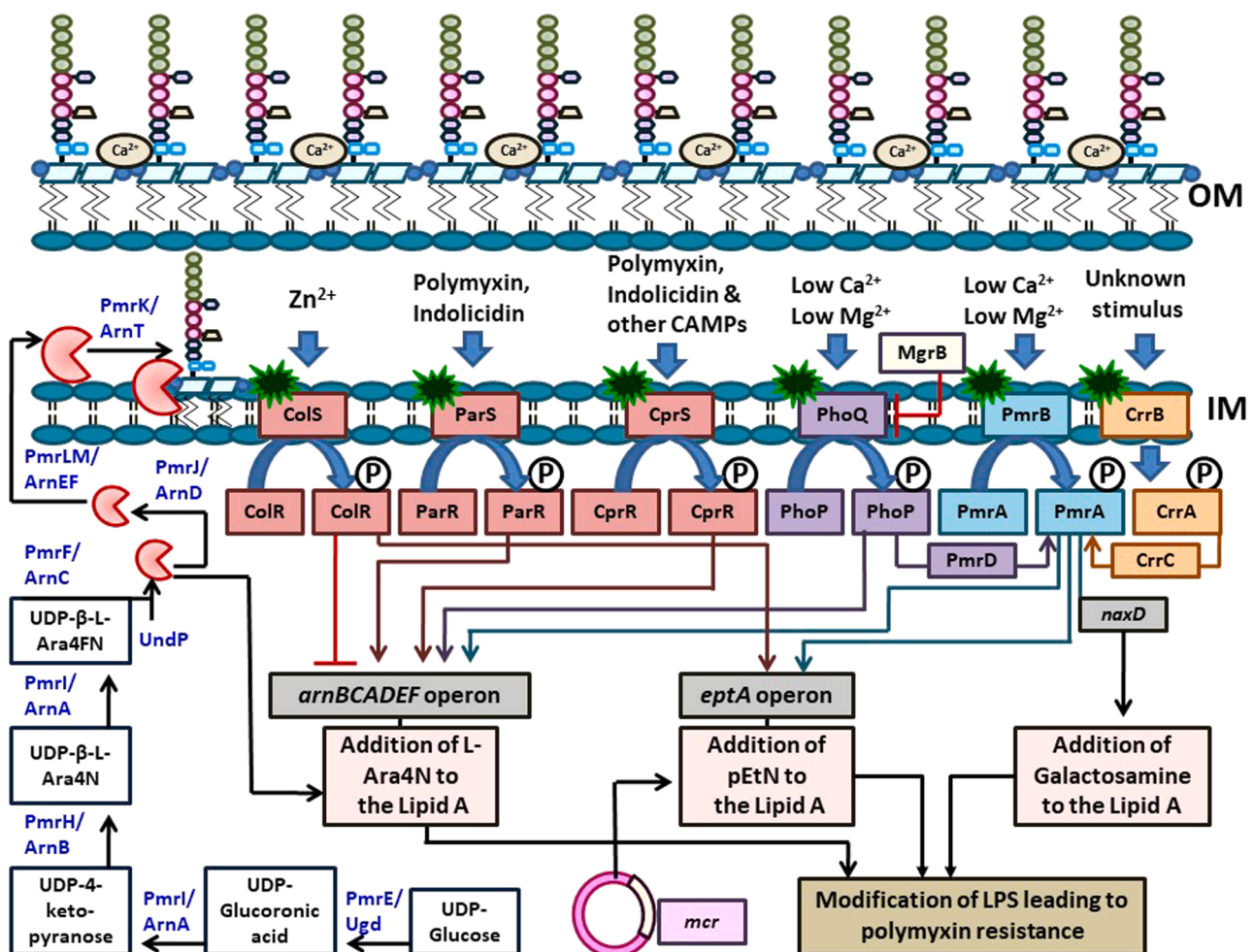


Fig. 4. Polymyxin resistance mechanisms in Gram-negative bacteria. The steps involved in the biosynthesis, transport, and addition of L-Ara4N to the lipid A component of LPS are depicted schematically. The activities of different TCS modules in the modification of the LPS are shown as well as the role of mobile colistin resistance genes is also depicted. The enzyme requirements for each step are indicated above or to the side of each arrow. *P. aeruginosa* and *K. pneumoniae* specific TCS are shown in brown and orange color respectively. PhoPQ TCS present in both is shown in purple, whereas PmrAB TCS found in *A. baumannii*, *P. aeruginosa*, and *K. pneumoniae* is shown in light blue color. (See the text for details).

subsequently the modification of LPS. Among Gram-negative bacterial species, two well-known TCSs with overlapping functions, PhoPQ and PmrAB, are primarily involved in LPS modification pathways. Environmental cues such as a decrease in Mg^{2+} and Ca^{2+} concentrations in the cell envelope, low pH, and the presence of CAMPs are sensed by the sensor kinases PhoQ and PmrB and affect the expression patterns of these TCSs via phosphorylation mediated mechanisms. Activated PhoQ and PmrB phosphorylate their cognate response regulators PhoP and PmrA, respectively, affecting their binding to the targeted genes and modulating their expression. The overexpression of the *pmrCAB* and the *arnBCADTEF* operon facilitates the addition of pEtN and L-Ara4N to LPS (Olaitan et al., 2014). Interestingly, in *E. coli*, PmrAB is activated by the PhoPQ system via a small protein PmrD that results in the LPS modification (Kox et al., 2000; Winfield and Groisman, 2004; Kato et al., 2012). The details of the pathways involved in the LPS modification mediated via TCSs are elaborated in Fig. 4.

The LPS modification pathways among the Gram-negative bacterial pathogens are largely conserved, albeit with subtle variations. In the following sections, the LPS modification pathways observed among some important nosocomial pathogens are described as well and the distinct mechanisms responsible for the polymyxin resistance in these organisms are summarized in Table 1.

5.1.1. *Acinetobacter baumannii*

A. baumannii is a Gram-negative opportunistic pathogen responsible for high mortality in critically sick individuals. Interestingly, only a single TCS PmrAB is involved in polymyxin resistance development via LPS modification in this organism. The addition of pEtN to the LPS generally occurs due to mutations in *pmrA* and *pmrB* genes as observed previously by Adams et al. (2009). Interestingly, mutations in *pmrAB* that downregulate the *pmrCAB* operon have been observed to revert the colistin-resistant *A. baumannii* to a susceptible phenotype (Snitkin et al., 2013; Cheah et al., 2016). Moreover, the downregulation of *pmrC* makes the colistin-resistant *A. baumannii* increasingly sensitive to it (Arroyo et al., 2011; Lima et al., 2018). These studies highlight the primacy of the PmrAB TCS in the development of polymyxin resistance in *A. baumannii*.

The L-Ara4N modification of the LPS has not been observed in *A. baumannii*; though, glycosylation of lipid A with galactosamine by NaxD deacetylase has been detected (Chin et al., 2015). The addition of galactosamine to the LPS molecule causes the masking of the negative charge on lipid A phosphate groups due to its structural similarity to L-Ara4N. The *naxD* is a PmrB-regulated gene indicating that the sensing of polymyxin by the bacterium triggers the glycosylation pathway mediated by the TCS (Chin et al., 2015). Clinical isolates having LPS modification with pEtN and galactosamine have also been observed in patients undergoing colistin treatment (Pelletier et al., 2013; Chin et al., 2015). These observations strongly indicate why the TCS-mediated LPS modification pathways need to be explored as novel drug targets.

Formation of OMVs in Gram-negative bacteria is a formidable strategy against the onslaught of antimicrobial agents. Recently, *A. baumannii* was shown to employ negatively charged OMVs to trap the cationic polymyxins in the culture medium and neutralize them (Park et al., 2021). Interestingly, the OMVs could help the polymyxin-susceptible strains survive the effect of polymyxins (Park et al., 2021). Although this is the first report of such a mechanism in *A. baumannii*, the prevalence of which in nature needs further investigation.

An intriguing pathway to polymyxin resistance in *A. baumannii* involves the total loss of the LPS molecule (Moffatt et al., 2010). The LPS deficient phenotype results from the mutations in genes encoding LpxA, LpxC, and LpxD that are involved in catalyzing the first three steps in the lipid A biosynthesis pathway (Fig. 2). Most significantly, nucleotide substitutions in *lpxA* (viz. C→G at nt 475, G→A at nt 203, and C→A at nt 214) and *lpxC* (viz. C→T at nt 89) are found to produce non-synonymous mutations in the LpxA and LpxC proteins respectively, affecting their

Table 1
Polymyxin resistance mechanisms employed by the prominent nosocomial pathogens.

Bacterial species	Polymyxin resistance mechanisms	Genes/ Operons Involved	References
<i>A. baumannii</i>	• TCS-mediated outer membrane remodelling/modification	<i>pmrA</i> , <i>pmrB</i>	(Adams et al., 2009)
	• Complete loss of LPS	<i>lpxA</i> , <i>lpxC</i> , <i>lpxD</i>	(Moffatt et al., 2010, 2011; Henry et al., 2012; Powers and Trent, 2018)
	• Efflux pump systems	<i>adeABC</i> , <i>HlyD</i> family, <i>emrA</i> , <i>emrB</i>	(Lin et al., 2017a, 2017b; Boinett et al., 2019)
	• Mobilization of <i>mcr</i> genes by plasmids and transposons	<i>mcr-1</i> , <i>mcr-2</i> , <i>mcr-3</i> , <i>mcr-4</i>	(Hameed et al., 2019; Al-Kadmy et al., 2020; Snyman et al., 2021)
<i>P. aeruginosa</i>	• TCS-mediated outer membrane remodelling/modification	<i>phoP</i> , <i>phoQ</i> , <i>pmrA</i> , <i>pmrB</i> , <i>parR</i> , <i>parS</i> , <i>colR</i> , <i>colS</i> , <i>cprR</i> , <i>cprS</i>	(Moskowitz et al., 2004, 2012; Miller et al., 2011; Fernández et al., 2012; Gutu et al., 2013; Nowicki et al., 2015)
	• Efflux mechanisms	PA4456, PA4455, PA4454, PA4453, PA4452, <i>oprD</i>	(Lorusso et al., 2022)
	• Loss of Lipid-A	<i>lpxC</i> , <i>lptD</i> , and <i>lpxO₂</i>	(Dosselmann et al., 2017; Sherry and Howden, 2018)
	• Defective biofilm formation	<i>ppk</i> , <i>modA</i>	(Azimi and Lari, 2019)
	• Mobilization of <i>mcr</i> genes by plasmids and transposons	<i>mcr-1</i> , <i>mcr-2</i> , <i>mcr-3</i> , <i>mcr-5</i> ,	(Tahmasebi et al., 2020; Chen et al., 2022)
<i>K. pneumoniae</i>	• TCS-mediated outer membrane remodelling/modification	<i>phoP</i> , <i>phoQ</i> , <i>pmrA</i> , <i>pmrB</i> , <i>pmrD</i> , <i>crrA</i> , <i>crrB</i> , <i>crrC</i> , <i>mgrB</i>	(Poirel et al., 2015; Wright et al., 2015; Jayol et al., 2017; McConville et al., 2020)
	• Capsular polysaccharide overexpression	<i>cps</i> operon, <i>siaD</i>	(Formosa et al., 2015)
	• Expression of efflux pumps and outer membrane porins	<i>acrAB</i> , <i>mdtK</i> , <i>tolC</i> , <i>kpnEF</i> , <i>yrbB-F</i> , <i>oqxAB</i> , <i>ompA</i>	(Srinivasan and Rajamohan, 2013; Olaitan et al., 2014; Singh et al., 2021)
	• Mobilization of <i>mcr</i> genes by plasmids and transposons	<i>mcr-1</i> , <i>mcr-3</i> , <i>mcr-7</i> , <i>mcr-8</i> ,	(Yang et al., 2018; Chen et al., 2021)

function. Moreover, large deletions in *lpxA* (viz. 445-bp deletion at nt 364, and 30-bp deletion at nt 391) and *lpxC* (viz. 84-bp deletion at nt 135) have also been identified in the polymyxin-resistant *A. baumannii* strains (Moffatt et al., 2010). The non-availability of the polymyxin binding site in these LPS-deficient strains leads to a very high degree of polymyxin resistance (Moffatt et al., 2010, 2011; Powers and Trent, 2018). Remarkably, *A. baumannii* can survive without the LPS, and it is most likely supplemented by an increased synthesis of lipoproteins, phospholipids, and the polysaccharide poly-1,6-N-acetylglucosamine (PNAG) (Henry et al., 2012). Furthermore, an entirely novel phenomenon was recently discovered in the LPS-deficient *A. baumannii* strains, where the growth depended on the presence of colistin in the medium (Lee et al., 2017; Zhu et al., 2020; Nang et al., 2021). In the Mueller Hinton agar containing colistin discs the colonies grew surrounding the disc and hardly any growth was observed farther from the disc. Interestingly, the authors have shown that these colistin-dependent

A. baumannii strains can develop a high level of resistance without the selection within a few generations of serial passage (Lee et al., 2017). On characterization, the LPS-deficient colistin-dependent strains showed extensive OM remodelling that may explain the basis of this phenomenon (Zhu et al., 2020). Though such polymyxin-dependent phenomena are yet to be discovered in other Gram-negative bacteria, the implications of such strains in the clinic cannot be ignored.

5.1.2. *Pseudomonas aeruginosa*

P. aeruginosa is a common nosocomial pathogen that can cause serious infections in people with compromised immune systems. To date, five TCSs namely PhoPQ (Macfarlane et al., 1999; Barrow and Kwon, 2009; Miller et al., 2011), PmrAB (McPhee et al., 2003; Moskowitz et al., 2004, 2012), ParRS (Fernández et al., 2010, 2012), CprRS (Gutu et al., 2013) and ColRS (Nowicki et al., 2015) have been identified to have a vital role in the polymyxin resistance development in *P. aeruginosa* (Fig. 4) (Table 1).

Like other Gram-negative pathogens, polymyxin resistance in *P. aeruginosa* primarily emerges by the addition of L-Ara4N to lipid A with the activation of the *arnBCADTEF* operon, by the PmrAB and PhoPQ TCSs on detecting various environmental perturbations such as low levels of Mg^{2+} or Ca^{2+} , and low pH. Studies have also revealed that mutations in the genes encoding regulatory proteins cause the constitutive expression of LPS modification factors (Miller et al., 2011; Olaitan et al., 2014; Perez et al., 2019). Interestingly, reversions of polymyxin-resistant to polymyxin-susceptible phenotype in *P. aeruginosa* can be caused by mutations in TCS, for example, mutations associated with *pmrAB*, or additional mutations in the *phoP*, or truncation in *phoQ*, etc. (Miller et al., 2011; Lee and Ko, 2014; Olaitan et al., 2014).

Unlike in the case of *E. coli*, where PmrD connects the TCSs PhoPQ and PmrAB, in *P. aeruginosa* phosphorylated PhoP directly activates PmrAB causing overexpression of *arnBCADTEF-ugd* operon and LPS modification (Macfarlane et al., 1999; Jeannot et al., 2017). An additional TCS named ColRS is also involved in adding phosphoethanolamine (pEtN) to lipid A in the presence of Zn^{2+} (Fig. 4) (Nowicki et al., 2015; Kaye et al., 2016). However, a recent report suggests that in *P. aeruginosa*, the LPS modification with pEtN has a limited role in developing polymyxin resistance in comparison to L-Ara4N modification as the *arn* and *eptA* genes involved in the respective pathways are triggered differentially in response to the antibiotic stress (Cervoni et al., 2023). Previous work has also shown that *P. aeruginosa* *lpxC*, *lptD*, and *lpxO₂* gene mutations impair lipid A synthesis and cause the loss of the polymyxin target. Although these mutations are often observed in *in vitro* evolutionary studies, they are rarely detected clinically, probably due to bacterial fitness issues (Dosselmann et al., 2017; Sherry and Howden, 2018).

5.1.3. *Klebsiella pneumoniae*

Given its widespread prevalence, *K. pneumoniae* is a notorious opportunistic pathogen, and three TCSs namely PhoPQ, PmrAB, and CrrAB are shown to be involved in polymyxin resistance development in this organism (Table 1) (Wright et al., 2015; McConville et al., 2020; Dodan et al., 2023; Lu et al., 2023a). The lipid A modification is a result of mutations in the genes *pmrA*, *pmrB*, or *phoQ* that lead to the overexpression of PhoPQ and PmrAB TCSs, resulting in the addition of L-Ara4N or pEtN to LPS (Olaitan et al., 2014). Moreover, a small transmembrane protein MgrB (47 aa) inhibiting the PhoQ kinase is also involved in this pathway, as its inactivation leads to the development of polymyxin resistance (Lippa and Goulian, 2009). MgrB inactivation causes constitutive PhoQ kinase activity leading to LPS remodelling resulting in polymyxin resistance (Cannatelli et al., 2013, 2014; Poirel et al., 2015; Bray et al., 2022).

In *K. pneumoniae* the phosphoethanolamine transferase, encoded by *eptB* is negatively regulated by PhoPQ via MgrB and is also involved in the LPS modification pathway. The activation of *eptB* results in the

addition of pEtN to the outer Kdo residues of the LPS. As a result, there is less repulsion between adjacent LPS molecules which leads to polymyxin resistance (El-Sayed Ahmed et al., 2020). Recently, mutations in the *crrAB* TCS have also been implicated in the colistin resistance pathway (Jayol et al., 2017; Sun et al., 2020). Inactivation of *crrB* causes overexpression of the *pmrAB* that in turn activates the *pmrHFLJKLM* operon, *pmrC*, and *pmrE* genes, resulting in the modification of lipid A with L-Ara4N and pEtN (Fig. 4) (Cheng et al., 2016; Sun et al., 2020).

5.2. Horizontally acquired polymyxin resistance

Even though Gram-negative pathogens frequently exhibit several chromosome-encoded polymyxin resistance mechanisms, the identification of “mobile colistin resistance (*mcr*)” genes in the environment in recent times and their extent of global spread have created a serious issue that is highlighted in the following paragraphs.

Horizontal gene transfer (HGT) events can also lead to acquiring polymyxin resistance. Primarily the plasmid-borne “mobile colistin resistance” (*mcr*) gene has become a prominent source of polymyxin resistance spreading among Gram-negative bacteria. The plasmid-mediated *mcr-1* gene was initially reported in 2015 from an *E. coli* strain isolated from food animals in China (Gao et al., 2016; Liu et al., 2016). The *mcr-1* was subsequently detected in a variety of *Enterobacteriaceae* species, including *E. coli*, *Klebsiella* spp., *Enterobacter* spp., *Salmonella* spp., *Shigella* spp., and *Cronobacter* spp. etc. (Poirel et al., 2017a; Mmatli et al., 2020, 2022; Lu et al., 2023b). Investigations have shown that more *mcr-1* positive isolates are from animal origin than from human sources leading to the hypothesis that *mcr-1* originated in animals, namely pigs, and cattle, and subsequently spread to humans (Aghapour et al., 2019; Nang et al., 2019; Gaballa et al., 2023).

As the *mcr-1* gene is carried on a conjugative plasmid, it can easily spread among bacteria of both animal and human origin. Though the *mcr-1* gene was initially discovered in 2015, retrospective surveillance of bacterial strains has shown their presence in samples isolated decades back (Nang et al., 2019). In fact, the earliest *mcr-1*-positive *E. coli* strains were detected in poultry samples from the 1980 s in China, which is coincident with the use of colistin in farming (Schwarz and Johnson, 2016; Shen et al., 2016). Subsequent to the discovery of *mcr-1* gene, ten different classes of *mcr* genes (*mcr-1* to *mcr-10*) including several variants have been identified among bacteria (Ayoub Moubareck, 2020; Mmatli et al., 2022). Fig. 5 presents a timeline of the discovery and detection of the individual classes of *mcr* genes as per their country of origin and the bacterial sources (Nang et al., 2019; Mmatli et al., 2022).

The *mcr* genes encode the enzyme pEtN transferase that catalyzes the addition of pEtN to lipid A resulting in the LPS modification, a mechanism similar to that is regulated by the TCS modules described in the previous sections. The chemical configuration of pEtN transferase is responsible for this activity (Gao et al., 2016; Sun et al., 2018; Gaballa et al., 2023). The MCRs pEtN transferases are localized in the cell in such a way that its N-terminal domain is present in the IM, whereas the C-terminal catalytic phosphoethanolamine transferase domain remains in the periplasm (Sun et al., 2017). This permits a pEtN moiety to be added to the outer Kdo residue of LPS as a result of phosphatidyl ethanolamine cleavage. The mechanism used by MCR enzymes to increase polymyxin resistance is like that seen in intrinsically resistant strains. A similar mechanism of LPS modification with pEtN is observed in resistant strains of *E. coli*, *K. pneumoniae*, and *A. baumannii* with mutations in the regulatory gene *ept* (Reynolds et al., 2005). Recent studies show that the expression of MCR-1 alone increases the colistin MICs in *E. coli* by 4- to 8-fold, in the absence of additional resistance mechanisms (Poirel et al., 2017a). Therefore, the acquisition of the *mcr* plasmids by bacterial pathogens has an alarming consequence for the healthcare sector.

The crystal structure of MCR-1, specifically its catalytic domain, was investigated by Stojanoski et al. (2016) revealing its structural homology with the phosphoethanolamine transferases. Subsequently, the crystal structure of the MCR-2 sharing significant homology with the

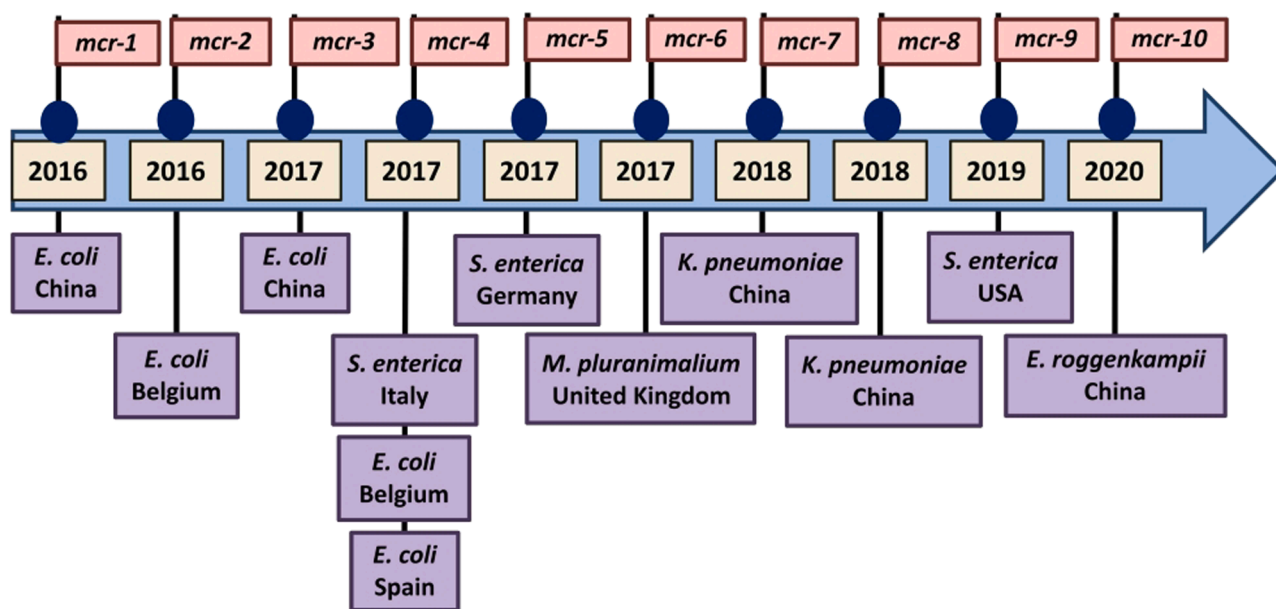


Fig. 5. The timeline of the first report of different classes of mobile colistin resistance (*mcr*) genes. The detection of *mcr-1* to *mcr-10* classes are depicted and their source bacterial species and the country of origin are also shown in the timeline.

MCR-1 sequence was also investigated (Coates et al., 2017). MCR-1's catalytic region contains amino acid residues that share sequence conservation with the enzymes LptA and EptC from *Neisseria meningitidis* and *Campylobacter jejuni*, respectively (Wanty et al., 2013; Fage et al., 2014; Xavier et al., 2016). As revealed by *in silico* analysis the amino acid sequence of MCR shows homology with the pEtN transferases (*pmrC*) identified in *Paenibacillus* spp., as well as other Gram-negative bacterial species, some of which are inherently resistant to colistin (Liu et al., 2016; Poirel et al., 2017b; Quiroga et al., 2019). As per the pieces of evidence presented above it is quite possible that the *mcr* plasmids initially originated among the colistin-producing bacterial species that have subsequently spread to other microorganisms.

Though the spread of *mcr* genes via plasmids is well established, it is also worth highlighting the role of transposable elements in mobilizing the *mcr* genes (Wang et al., 2018; Goodman et al., 2023). Genetic analysis of various *mcr* plasmids has revealed the presence of several IS elements in the *mcr* gene neighborhood aiding in their transposition and spread. Transposons such as Tn6330 containing two copies of the IS element IS*ApII* are primarily implicated in the transposition of the *mcr* genes (Li et al., 2017; Poirel et al., 2017a; Wang et al., 2018; Nang et al., 2019). The major benefit provided by such elements is the transposition of the *mcr* genes to different mobile genetic elements (MGEs) and helping in their rapid dissemination among the bacterial species by horizontal gene transfer (HGT) mechanisms (Gao et al., 2016; Li et al., 2018; Goodman et al., 2023). The role of *mcr* genes in the spread of polymyxin resistance has been extensively reviewed by Nang et al. (2019) and Mmalti et al. (2022) recently.

The burden of *mcr* and its widespread dissemination has alarmed the global community. Investigation into the molecular epidemiology of the *mcr* genes and the resistance mechanisms encoded by them may assist in designing therapeutic interventions against severe infections caused by pathogens carrying *mcr* plasmids. The risk of uncontrolled transmission is apparent, and urgent interventions must be in place to check the resistance spreading further.

6. Molecular interactions of polymyxin with the LPS complex

As previously stated, knowledge of polymyxin-LPS interactions is required for a greater understanding of the polymyxins' mechanisms of action and resistance, and their structure-activity relationship (SAR).

Understanding the mechanism and dynamics of these phenomena is essential for creating polymyxin analogs that are efficient against polymyxin-resistant strains. The binding between polymyxin and LPS has been investigated using a host of biophysical methods, including isothermal titration calorimetry (ITC) (Srima et al., 1996; Brandenburg et al., 2005; Howe et al., 2007), fluorescence probe displacement (David et al., 1992; Thomas et al., 1998), surface plasmon resonance (SPR) (Thomas and Surlia, 1999; Thomas et al., 1999), nuclear magnetic resonance (NMR) spectroscopy (Martin et al., 2003; Mares et al., 2009; Velkov et al., 2010), and neutron reflectometry (Paracini et al., 2018), etc. Some of the pivotal studies conducted in this area are presented in the following sections.

6.1. Polymyxin analogs and their interaction with the LPS

The interaction between polymyxin and LPS is an important part of its antibiotic activity, therefore, efforts have been made to develop polymyxin analogs with comparable affinity for LPS and lesser nephrotoxicity. Moreover, developing polymyxin derivatives having synergistic effects with other antibiotics against Gram-negative bacterial infections is also an area of active research. The novel polymyxin derivatives are synthesized using two approaches, (i) the total synthesis approach based on modifications at specific sites within the polymyxin's core heptapeptide through solid-phase peptide synthesis; and (ii) the semi-synthetic approach based on an appropriate protection and deprotection strategy where modification occurs particularly around linear tripeptide and/or N-terminus region (Brown and Dawson, 2017).

Previously Kanazawa et al. (2009) had revealed the role of each amino acid present in the polymyxin structure, specifically the importance of the Dab residues by developing various polymyxin derivatives using alanine scanning. The study showed that the location of the Dab residues in the cyclic heptapeptide ring is more crucial for its bactericidal activity than in the linear tripeptide (Kanazawa et al., 2009). Furthermore, the N-terminal fatty acid moiety is critical for the bactericidal activity of polymyxin, which provides a major driving force for the polymyxin-LPS interaction, whereas the heptapeptide cyclic core helps to permeabilize the OM (Kimura et al., 1992; Srima et al., 1996). Several research groups have been pioneering the research effort to develop the next generation polymyxins based on these early findings (Vaara, 2013, 2019a; Velkov et al., 2013; Rabanal and Cajal, 2017).

Polymyxin analogs that have been developed in the preceding decades have been extensively reviewed in recent publications (Nang et al., 2021; Chiu et al., 2022) as well as summarized in Table 2. A compound named CB-182,804 was the first polymyxin derivative to be clinically tested under FDA supervision, however, was discontinued after the phase 1 clinical trial in 2010 because of the toxicity issues (Quale et al., 2012). Polymyxin analogs lacking antibacterial activity, nevertheless involved in permeabilizing the OM have been developed as “permeabilizers” or “sensitizers” and are supposed to be used in combination with other bactericidal antibiotics (Vaara, 2019b; Nang et al., 2021). In this context, notable permeabilizers such as PMBN (polymyxin B nonapeptide), NAB7061, SPR741, SPR206, MRX-8, and QPX9003 have completed the phase I clinical trials (Vaara, 2019b; Nang et al., 2021; Aslan et al., 2022; Prasad et al., 2022).

Efforts are also being made to develop polymyxin analogs with reduced toxicity by tweaking the amino acid composition and changing their positions in the polymyxin scaffold. Slingerland et al. (2022) using a semisynthetic approach modified the linear tripeptide part of the polymyxin structure and developed compounds with reduced nephrotoxic and comparable antibacterial activity. This study specifically used a reductive disulfide linkage at the N-terminal end of the molecule

which has shown promising results using human renal epithelial cells (Slingerland et al., 2022). More recently, polymyxin analogs containing unnatural amino acids in the structure have been developed and assessed for their potential antibacterial activity (Jorgensen et al., 2024). Using halogenated residues and other unnatural amino acids in the hydrophobic part of the molecule, analogs have been developed with improved bactericidal activity even against colistin-resistant bacterial strains (Jorgensen et al., 2024). Though the preliminary toxicity assessment has shown promising results against the mammalian erythrocytes, the full toxicity assessment and other pharmacokinetics studies remain to be performed.

Polymyxin analogs have also been used in various biophysical experiments to get a deeper understanding of the polymyxin-LPS interactions. In this context, Dansyl-polymyxin B, a fluorescent derivative of polymyxin B that is generally used to study the interaction of the drug with the LPS as well as its pharmacokinetics resulted in a 20-fold increase in fluorescence (Moore et al., 1986). This study used the fluorescent probe displacement test as a surrogate for analyzing polymyxin-LPS structure-activity interactions. While the fluorescence of the dansyl molecule is minimal in polar environments, it is greatly enhanced when interacting with hydrophobic structures in the LPS

Table 2

List of prominent polymyxin analogs, their characteristic features, and their clinical development status.

Compound	Modification in comparison to Polymyxins	Effectiveness	Development status	References
FADDI-02	FA: Methyl-octanoyl/ methyl-heptanoyl → Octanoyl Leu ⁷ → Amino-decanoyl (Ada)	Greater antimicrobial activity than PMB against Pol ^R strains	N/A	(Velkov et al., 2014)
QPX9003	FA: Methyl-octanoyl/ methyl-heptanoyl → 2,4-dichlorobenzoyl Dab ³ → Diamino- propinoyl (Dap ³) -Phe ⁶ → D-Leu Leu ⁷ → L-2-aminobutyric acid (Abu)	Superior efficacy and less toxicity in mouse and rat pneumonia models in comparison to PMB	Phase I clinical trial completed in 2022	(Sabet et al., 2019; Roberts et al., 2022)
NAB739	FA: Methyl-octanoyl/ methyl-heptanoyl → Octanoyl No Dab at AA ¹ Dab ² → Thr Dab ³ → D-Ser	Similar activity against <i>E. coli</i> (MIC ₉₀ -1–2 mg/l) and slightly higher in <i>K. pneumonia</i> (2-fold) but weaker activity against <i>P. aeruginosa</i> (MIC ₉₀ . 8–16 mg/l) with decreased cytotoxicity (CC ₅₀ - 337 mg/L versus 13 mg/L for polymyxin) in HK-2 cell line	N/A	(Vaara et al., 2013; Vaara and Vaara, 2013)
Pfizer 5X	FA: Methyl-octanoyl/ methyl-heptanoyl → 6-oxo-1-phenyl-1,6-dihydropyridine-3-carbonyl Dab ³ → Di-amino-propinoyl (Dap ³)	Similar activity against susceptible strains to polymyxin with less cytotoxicity in the hRPTEC cell line with TC ₅₀ of > 100 μM versus 22 μM for polymyxin	N/A	(Magee et al., 2013)
CB-182 804	FA: Methyl-octanoyl/ methylheptanoyl → 2-chloro-phenylamino-carbonyl	Identical MIC activity to PMB and reduced cytotoxicity in LLC-PK1 cell line (EC ₅₀ > 1000 mg/ml versus 318 mg/ml for PMB)	Phase I clinical trial completed	(Quale et al., 2012)
CA824	FA: Methyl-octanoyl/ methyl-heptanoyl → (S)-1-(2-methylpropyl)-piperazine-2-carbonyl ⁺ No Dab at AA ¹	Similar or slightly improved MIC activity against <i>E. coli</i> , <i>K. pneumonia</i> , and <i>P. aeruginosa</i> to PMB and reduced cytotoxicity in HK-2 cell line (TC ₅₀ 148 μg/ml versus 15 μg/ml for PMB)	N/A	(Brown and Dawson, 2017; Vaara, 2019a)
SPR206	Alteration of fatty acyl tail with short linear segment FA: Methyl-octanoyl/ methyl-heptanoyl → (3 S)-4-amino-3-(3-chlorophenyl) butanoyl Dab ¹ → No Dab	Lower MICs to polymyxin against MDR organisms and reduced cytotoxicity in HK-2 cell line as well as in mouse models	Phase I clinical trial completed	(Vaara, 2019b; Zhang et al., 2020; Bruss et al., 2021)
MRX-8	Fatty acyl tail linked via an ester bond FA: Methyl-octanoyl/ methyl-heptanoyl → 3-(2,2-Dimethyl-butanoyloxy)-propanoyl	Higher MICs to polymyxin but better efficacy against lung infections caused by Gram-negative organisms in mice	Phase I	(Duncan et al., 2022; Wu et al., 2022)
NAB741/ SPR741	FA: Methyl-octanoyl/ methyl-heptanoyl → Acetyl Dab ¹ → No Dab Dab ³ → D-Ser	Higher MIC activity against some GNB (≥ 32 μg/ml). Renal clearance 400-fold higher than colistin in the rat model and reduced renal toxicity in the Cynomolgus monkey model	Phase I clinical trial completed	(Vaara et al., 2010; Vaara and Vaara, 2010; Corbett et al., 2017)
PMBN	FA: Methyl-octanoyl/methyl-heptanoyl → No Fatty acyl chain Dab ¹ → Deleted	Higher MIC activity against <i>E. coli</i> and <i>K. pneumoniae</i> (> 500 mg/L) but the low value in the case of <i>P. aeruginosa</i> (8 mg/L) with less cytotoxicity in the HK-2 cell line with TC ₅₀ of > 1000 μM	Phase I clinical trial completed	(Vaara and Vaara, 1983; Vaara, 1992; Keirstead et al., 2014)
Compound 38	FA: Methyl-octanoyl/ methyl-heptanoyl → Octanoyl Dab ³ → Gly	Effective MIC activity (0.5–2 μg/ml) with reduced cytotoxicity in HepG2 and HEK293 cell lines (CC ₅₀ > 300 μM)	N/A	(Gallardo-Godoy et al., 2016; Velkov et al., 2016)

FA: Fatty acid, PMB: polymyxin B, AA: amino acid residue, Pol^R: polymyxin resistant bacteria, N/A: not available

(Moore et al., 1986). The first completely synthetic mono-dansylated polymyxin probe [dansyl-Lys]¹polymyxin B₃ (DPmB3), where one of the six Dab residues found in polymyxin was swapped with the dansyl moiety, was utilized to explore the polymyxin-LPS interaction in the subsequent experiments (Soon et al., 2011). The DPmB3 probe showed an increased affinity for the purified LPS molecule in comparison to the whole cell and has been useful in further investigating the SAR inside the polymyxin scaffold. This derivative has the potential to be used for drug screening assays (Soon et al., 2011).

A synthetic probe that pharmacologically mimics polymyxin B was developed by attaching the dansyl group to an octyl glycine residue and incorporating it into the polymyxin scaffold's N-terminus (Deris et al., 2014b). While this mimics the N-terminal fatty acyl groups in polymyxins, the rest of the scaffold's essential properties are preserved in this probe. This may be regarded as an improved version of the polymyxin probes for fluorescence displacement tests as well as fluorescence microscopic studies and provide a better platform to discover new polymyxin analogs that are effective against resistant strains. Another polymyxin B derivative that has been successfully used in the fluorescence displacement, as well as fluorescence microscopic studies, is MIPS-9451 which has provided quantitative information on the polymyxin-LPS interactions and determined the binding affinities of MIPS-9451 with LPS from a range of bacterial species (McInerney et al., 2016). An improved strategy to design a novel chimeric peptidomimetic antibiotic (CPA) combining the polymyxin B nonapeptide (PMBN) with macrocyclic β -hairpin motif (MHM) was attempted recently (Javed et al., 2023), that showed superior bactericidal and LPS neutralizing ability. The CPA could inhibit macrophage activation and the inflammatory effects in comparison with conventional antibiotics against Gram-negative bacteria (Javed et al., 2023).

These novel polymyxin derivatives not only have provided deeper insights into the molecular interaction between the LPS and the drug but also have the potential to be utilized in drug screening efforts and pharmacokinetics studies.

6.2. Structural studies of the polymyxin-LPS interactions

The molecular structure of the polymyxin-LPS interaction has also been investigated using NMR spectroscopy. The binding and structural data from such studies indicate that the polymyxin-LPS complex is stabilized by a mixture of two binding forces. One is electrostatic interaction that occurs between polymyxin's positive charges and the negative LPS head groups, and the other one is hydrophobic interaction occurring between each molecule's hydrophobic domains with the help of which polymyxin perforates the OM (Velkov et al., 2010, 2013). As per these models, polymyxin separates the polar and hydrophobic domains that are necessary for its antimicrobial activity by developing an envelope-like fold (Mares et al., 2009; Velkov et al., 2013). Recently, Goode et al. (2021) explored the molecular contact between polymyxin B and rough LPS (rLPS) inside the lipid membranes with the help of fluorescence dequenching methods. Solid-state NMR was used to assess how the dynamics and stability of lipid membranes change in response to polymyxin. The study revealed that the presence of the specific chemical types of LPS that are inherent to a bacterial species would influence the level of interaction with the polymyxin molecule and consequently the membrane lysis. Jiang et al. (2021a) have used an asymmetric OM model with rLPS to study the thermodynamics of polymyxin penetration into the OM. Employing coarse-grained (CG) simulations and free energy analysis this study revealed that the distribution of Ca²⁺ ions in the inner core of the LPS holds the structure intact thereby preventing the entry of polymyxins. However, an ionic imbalance in the core region is the prevalent mechanism of polymyxin-induced cell death. The role of Ca²⁺ homeostasis in response to antibiotics has also been explored using a machine-learning framework (Meyer et al., 2021). This study used Ca²⁺ as a hallmark of antibiotic response by integrating machine learning with Ca²⁺ activity

measurement within single cells. A variety of antibacterial substances were tested using this approach. Ca²⁺ dynamics perform an unrecognized involvement in membrane instability during polymyxin B treatment, and aberrant Ca²⁺ dynamics are observed preceding the membrane permeabilization as a result of polymyxin treatment. The period between these aberrant dynamics and membrane breakdown was regulated by membrane voltage and external Ca²⁺ concentration (Meyer et al., 2021). All these studies highlight the role of membrane dynamics being perturbed in the presence of polymyxins and such techniques can be adapted to study the distinct membrane physiology in the development of resistance at a single-cell level.

6.3. Molecular dynamics simulation studies of LPS-polymyxin interactions

Further insights into the polymyxin-LPS interaction have also come from several molecular dynamics (MD) simulation studies, which predict the behavior of a system when it is perturbed by some agents (Weerakoon et al., 2021). Using coarse-grained atomistic models of the bacterial OM the MD simulation studies have greatly complemented the data generated from biophysical experiments highlighting the importance of such approaches to understanding the molecular details of the polymyxin-LPS interactions (Jiang et al., 2021a; Weerakoon et al., 2021; Fu et al., 2022; Sun et al., 2022). Based on the MD simulation studies Fu et al., (2022) have proposed a model named SST for lipid Scramble, membrane phase Separation, and peptide Translocation, to provide insight into the polymyxin-OM interactions. The model suggests that the binding of polymyxin B leads to the loosening of LPS packing in the outer leaflet of the OM, followed by the development of membrane curvature deformities. To compensate for the deformities, phospholipids from the inner leaflet flip over to the outer leaflet causing the lipid scramble. This process leads to a phase separation of LPS and lipid void regions in the membrane. Meanwhile, the bound polymyxins gain entry initially into the periplasmic space and then to the LPS retained in the IM. The polymyxin then disrupts the IM in a similar process leading to cell lysis (Fu et al., 2022). Though some of the key steps proposed in the model are experimentally validated and look quite plausible, evidence for the whole model still needs to be worked out.

MD simulation studies have also been used in developing novel varieties of polymyxins with better membrane penetration resulting in higher effectiveness (Jiang et al., 2021b, 2022; Roberts et al., 2022). Jiang et al. (2021b) have systematically assessed the alanine substituted polymyxin analogs with the *A. baumannii* OM at the atomic level SAR modelling and the results have shown a good match between *in silico* and *in vitro* studies highlighting the potential of such approaches. Jiang et al. (2022) in their study using the MD simulation approach have taken into consideration the varieties of OM modifications in *P. aeruginosa* and its interactions with polymyxin based on the thermodynamic considerations. Combining computational findings with experimental data, this study has revealed the atomistic details of the polymyxin-OM interaction in the development of resistance. Using the chemical biology approach Roberts et al. (2022) have developed a compound named F365 having better efficacy against MDR pathogens such as *A. baumannii*, *P. aeruginosa*, and *K. pneumoniae*. Specifically, the non-conserved amino acid residues in the polymyxin scaffold have been tweaked and optimized to delineate its toxicity from its therapeutic efficacy showing promising results. These studies will have an appreciable influence on the development of polymyxin substitutes with better pharmacological properties in the future as more knowledge about the Gram-negative OM becomes available and modelling techniques are improved.

7. Concluding remarks

The increasing clinical use of polymyxins as a last-option antibiotic in the previous decades has resulted in the rising incidence of polymyxin resistance. This is further compounded by the use of polymyxins in

animal feeds as growth promoters that have driven the emergence and spread of *mcr* genes across the globe. Though several countries in recent years have restricted the use of various antimicrobial agents in animal feeds, its implications will be understood a few years down the line. A complete understanding of polymyxin's molecular interaction with the bacterial OM is necessary, as it is essential for its antibacterial activity. This interaction influences the membrane remodelling required for the development of polymyxin resistance. Moreover, the polymyxin-LPS interactions studies have provided critical insights to develop novel derivatives and polymyxin analogs such as PMBN, NAB7061, QPX9003, etc. that can be used as adjuvants in combination with other antibiotics. It is also crucial to conduct more pharmacokinetics/ pharmacodynamics (PK/PD) studies to assess the safety and effectiveness of polymyxin and its novel analogs as a mono- or combination therapeutic option (MacNair et al., 2018; Carfrae et al., 2023). The diminishing drug discovery pipeline combined with the fast-rising nosocomial infections during the COVID-19 pandemic has made these investigations even more necessary. Improvement in the biophysical analytical techniques supplemented with a host of MD simulation studies has contributed to this direction. These studies have shown promising results in developing better versions/ analogs of polymyxins and other CAMPs that have improved penetrance and stability in the bacterial membrane. Interestingly, few studies indicate the ability of polymyxin to bind ribosomes and hinder cell division. These modes of action of polymyxins need further clarity. Though bacteria can always develop resistance to antibiotics owing to their inherent nature, we need to be a few steps ahead against drug-resistant pathogens through research and innovation.

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CRediT authorship contribution statement

Saswat S. Mohapatra: Writing – review & editing, Writing – original draft, Supervision, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Sambit K. Dwibedy:** Writing – review & editing, Data curation. **Indira Padhy:** Writing – review & editing, Writing – original draft, Investigation, Data curation, Conceptualization.

Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Data Availability

No data was used for the research described in the article.

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